

1 Tables for depth $P = 1$

	graph_type depth	ER 1	HH 1	L2F 1	RR 1
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	-
	F_MPS.json	10 [40]	31 [39–144]	20 [40]	41 [40–100]
	L_MPS.json	10 [40]	31 [39–144]	20 [40]	41 [40–100]
	RTS_MPS.json	-	1 [106]	-	2 [40–100]
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	-	-	-
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	-
	F_PP.json	10 [40]	31 [39–144]	20 [40]	41 [40–100]
	L_PP.json	10 [40]	31 [39–144]	20 [40]	41 [40–100]
	RTS_PP.json	-	1 [106]	-	2 [40–100]
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	-	-	-
SV	FA_SV_noOpt.json	-	-	-	60 [10–20]
	FA_SV_opt.json	-	-	-	60 [10–20]
	F_SV.json	-	-	-	1 [10]
	L_SV.json	32 [10–17]	10 [12]	-	177 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	-	-	-	60 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	164 [10–20]

Table 1: **Number of Instances (num. nodes in brackets) for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 1	HH 1	L2F 1	RR 1
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	-
	F_MPS.json	0.8111 ± 0.0121	0.7542 ± 0.0118	0.7132 ± 0.0189	0.7697 ± 0.0153
	I_MPS.json	0.8111 ± 0.0121	0.7542 ± 0.0118	0.7132 ± 0.0189	0.7697 ± 0.0153
	RTS_MPS.json	-	0.744	-	0.75170 ± 0.00329
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	-	-	-
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	-
	F_PP.json	0.8111 ± 0.0121	0.7542 ± 0.0118	0.7132 ± 0.0189	0.7697 ± 0.0153
	I_PP.json	0.8111 ± 0.0121	0.7542 ± 0.0118	0.7132 ± 0.0189	0.7697 ± 0.0153
	RTS_PP.json	-	0.744	-	0.75170 ± 0.00329
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	-	-	-
SV	FA_SV_noOpt.json	-	-	-	0.7801 ± 0.0202
	FA_SV_opt.json	-	-	-	0.7813 ± 0.0206
	F_SV.json	-	-	-	0.774
	I_SV.json	0.7868 ± 0.0328	0.7929 ± 0.0297	-	0.8122 ± 0.0432
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	-	-	-	0.7813 ± 0.0206
	TS_SV.json	0.7868 ± 0.0328	0.7929 ± 0.0297	-	0.8145 ± 0.0441

Table 2: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 1$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

2 Tables for depth $P = 2$

	graph_type depth	ER 2	HH 2	L2F 2	RR 2
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	30 [40–70]
	F_MPS.json	10 [40]	31 [39–144]	20 [40]	41 [40–100]
	L_MPS.json	10 [40]	31 [39–144]	20 [40]	41 [40–100]
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	10 [40]	10 [39]	20 [40]	40 [40–70]
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	30 [40–70]
	F_PP.json	10 [40]	31 [39–144]	20 [40]	41 [40–100]
	L_PP.json	10 [40]	31 [39–144]	20 [40]	41 [40–100]
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	10 [40]	10 [39]	20 [40]	40 [40–70]
SV	FA_SV_noOpt.json	26 [10–17]	-	-	133 [10–20]
	FA_SV_opt.json	26 [10–17]	-	-	133 [10–20]
	F_SV.json	-	-	-	1 [10]
	L_SV.json	32 [10–17]	10 [12]	-	166 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	10 [12]	-	133 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	132 [10–20]

Table 3: **Number of Instances (num. nodes in brackets) for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 2	HH 2	L2F 2	RR 2
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	0.7567 ± 0.0371
	F_MPS.json	0.6930 ± 0.0126	0.7213 ± 0.0827	0.7334 ± 0.0311	0.7432 ± 0.0429
	I_MPS.json	0.78537 ± 0.00799	0.7591 ± 0.0615	0.7649 ± 0.0251	0.7504 ± 0.0423
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	0.77689 ± 0.00957	0.8307 ± 0.0157	0.7649 ± 0.0250	0.7637 ± 0.0358
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	0.8364 ± 0.0131
	F_PP.json	0.8583 ± 0.0251	0.8076 ± 0.0217	0.7890 ± 0.0216	0.8038 ± 0.0252
	I_PP.json	0.8712 ± 0.0126	0.8272 ± 0.0123	0.7903 ± 0.0187	0.8384 ± 0.0137
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	0.8712 ± 0.0126	0.8313 ± 0.0161	0.7903 ± 0.0187	0.8386 ± 0.0138
SV	FA_SV_noOpt.json	0.8400 ± 0.0240	-	-	0.8658 ± 0.0298
	FA_SV_opt.json	0.8544 ± 0.0262	-	-	0.8736 ± 0.0329
	F_SV.json	-	-	-	0.828
	I_SV.json	0.8537 ± 0.0265	0.8555 ± 0.0177	-	0.8556 ± 0.0509
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.8555 ± 0.0263	0.8664 ± 0.0247	-	0.8736 ± 0.0332
	TS_SV.json	0.8549 ± 0.0269	0.8708 ± 0.0188	-	0.8482 ± 0.0445

Table 4: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 2$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

3 Tables for depth $P = 3$

	graph_type depth	ER 3	HH 3	L2F 3	RR 3
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	10 [70]
	F_MPS.json	10 [40]	31 [39–144]	20 [40]	31 [40–100]
	L_MPS.json	10 [40]	31 [39–144]	20 [40]	31 [40–100]
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	-	10 [40]	10 [70]
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	10 [70]
	F_PP.json	10 [40]	31 [39–144]	20 [40]	31 [40–100]
	L_PP.json	10 [40]	31 [39–144]	20 [40]	31 [40–100]
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	-	10 [40]	10 [70]
SV	FA_SV_noOpt.json	26 [10–17]	-	-	132 [10–20]
	FA_SV_opt.json	26 [10–17]	-	-	132 [10–20]
	F_SV.json	-	-	-	1 [10]
	L_SV.json	32 [10–17]	10 [12]	-	166 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	10 [12]	-	132 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	131 [10–20]

Table 5: **Number of Instances (num. nodes in brackets) for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 3	HH 3	L2F 3	RR 3
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	0.7640 ± 0.0102
	F_MPS.json	0.8578 ± 0.0693	0.7558 ± 0.0682	0.7893 ± 0.0531	0.7654 ± 0.0507
	I_MPS.json	0.8859 ± 0.0197	0.7901 ± 0.0699	0.8487 ± 0.0188	0.8053 ± 0.0716
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	-	0.8885 ± 0.0108	0.7304 ± 0.0125
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	0.86971 ± 0.00950
	F_PP.json	0.8794 ± 0.0150	0.8388 ± 0.0272	0.8010 ± 0.0188	0.8466 ± 0.0191
	I_PP.json	0.9023 ± 0.0135	0.8673 ± 0.0111	0.8310 ± 0.0192	0.8724 ± 0.0149
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	-	0.7972 ± 0.0240	0.86963 ± 0.00952
SV	FA_SV_noOpt.json	0.8694 ± 0.0202	-	-	0.8966 ± 0.0219
	FA_SV_opt.json	0.8920 ± 0.0220	-	-	0.9105 ± 0.0268
	F_SV.json	-	-	-	0.874
	I_SV.json	0.8903 ± 0.0227	0.8976 ± 0.0148	-	0.8897 ± 0.0443
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.8783 ± 0.0222	0.8821 ± 0.0198	-	0.8987 ± 0.0261
	TS_SV.json	0.8922 ± 0.0248	0.9008 ± 0.0182	-	0.8741 ± 0.0404

Table 6: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 3$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

4 Tables for depth $P = 4$

	graph_type depth	ER 4	HH 4	L2F 4	RR 4
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	10 [40]
	F_MPS.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	L_MPS.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	10 [40]	10 [39]	20 [40]	20 [40]
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	10 [40]
	F_PP.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	L_PP.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	10 [40]	10 [39]	20 [40]	20 [40]
SV	FA_SV_noOpt.json	24 [10–17]	-	-	73 [10–20]
	FA_SV_opt.json	24 [10–17]	-	-	73 [10–20]
	F_SV.json	-	-	-	1 [10]
	L_SV.json	32 [10–17]	10 [12]	-	166 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	10 [12]	-	133 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	132 [10–20]

Table 7: **Number of Instances (num. nodes in brackets) for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

graph.type depth		ER 4	HH 4	L2F 4	RR 4
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	$0.925\,85 \pm 0.008\,14$
	F_MPS.json	0.7807 ± 0.0767	0.7595 ± 0.0820	0.8218 ± 0.0773	0.8788 ± 0.0594
	I_MPS.json	0.9280 ± 0.0147	0.7946 ± 0.0863	0.9008 ± 0.0139	0.9036 ± 0.0596
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	0.8436 ± 0.0590	0.8690 ± 0.0110	0.8944 ± 0.0147	0.8808 ± 0.0341
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	0.8992 ± 0.0156
	F_PP.json	0.8869 ± 0.0265	0.8643 ± 0.0275	0.8007 ± 0.0253	0.8744 ± 0.0192
	I_PP.json	0.9219 ± 0.0138	0.8932 ± 0.0106	0.8565 ± 0.0200	0.8931 ± 0.0163
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	0.9070 ± 0.0136	0.8728 ± 0.0126	0.8340 ± 0.0191	0.8797 ± 0.0150
SV	FA_SV_noOpt.json	0.8857 ± 0.0244	-	-	0.9263 ± 0.0213
	FA_SV_opt.json	0.9158 ± 0.0190	-	-	0.9313 ± 0.0213
	F_SV.json	-	-	-	0.913
	I_SV.json	0.9148 ± 0.0194	0.9177 ± 0.0149	-	0.9014 ± 0.0530
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.8971 ± 0.0211	0.8952 ± 0.0191	-	0.9114 ± 0.0252
	TS_SV.json	0.9165 ± 0.0222	0.9140 ± 0.0167	-	0.8921 ± 0.0364

Table 8: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 4$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

5 Tables for depth $P = 5$

	graph_type depth	ER 5	HH 5	L2F 5	RR 5
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	-
	F_MPS.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	L_MPS.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	RTS_MPS.json	-	1 [106]	-	2 [40–100]
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	20 [105–144]	10 [40]	-
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	-
	F_PP.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	L_PP.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	RTS_PP.json	-	1 [106]	-	2 [40–100]
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	20 [105–144]	10 [40]	-
SV	FA_SV_noOpt.json	20 [10–17]	-	-	63 [10–20]
	FA_SV_opt.json	20 [10–17]	-	-	63 [10–20]
	F_SV.json	-	-	-	1 [10]
	L_SV.json	32 [10–17]	10 [12]	-	166 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	10 [12]	-	132 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	132 [10–20]

Table 9: **Number of Instances (num. nodes in brackets) for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 5	HH 5	L2F 5	RR 5
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	-
	F_MPS.json	0.7947 ± 0.0818	0.7705 ± 0.0890	0.8611 ± 0.0834	0.8929 ± 0.0819
	I_MPS.json	0.95371 ± 0.00817	0.8152 ± 0.0873	0.9279 ± 0.0110	0.9321 ± 0.0363
	RTS_MPS.json	-	0.711	-	0.8373 ± 0.0729
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	0.7516 ± 0.0484	0.91392 ± 0.00952	-
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	-
	F_PP.json	0.9094 ± 0.0135	0.8842 ± 0.0265	0.8171 ± 0.0282	0.8898 ± 0.0153
	I_PP.json	0.9350 ± 0.0138	0.91053 ± 0.00968	0.8728 ± 0.0204	0.9054 ± 0.0178
	RTS_PP.json	-	0.882	-	0.8897 ± 0.0226
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	0.88187 ± 0.00789	0.8354 ± 0.0196	-
SV	FA_SV_noOpt.json	0.8966 ± 0.0256	-	-	0.9468 ± 0.0178
	FA_SV_opt.json	0.9340 ± 0.0173	-	-	0.9517 ± 0.0181
	F_SV.json	-	-	-	0.931
	I_SV.json	0.9320 ± 0.0175	0.92451 ± 0.00869	-	0.9160 ± 0.0494
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.9185 ± 0.0166	0.9071 ± 0.0202	-	0.9317 ± 0.0219
	TS_SV.json	0.9328 ± 0.0200	0.9262 ± 0.0132	-	0.9034 ± 0.0351

Table 10: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 5$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

6 Tables for depth $P = 6$

	graph_type depth	ER 6	HH 6	L2F 6	RR 6
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	10 [40]
	F_MPS.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	L_MPS.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	10 [40]	10 [39]	20 [40]	20 [40]
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	10 [40]
	F_PP.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	L_PP.json	10 [40]	31 [39–144]	20 [40]	21 [40–100]
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	10 [40]	10 [39]	20 [40]	20 [40]
SV	FA_SV_noOpt.json	11 [11–17]	-	-	43 [10–20]
	FA_SV_opt.json	11 [11–17]	-	-	43 [10–20]
	F_SV.json	-	-	-	-
	L_SV.json	32 [10–17]	10 [12]	-	165 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	10 [12]	-	132 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	132 [10–20]

Table 11: **Number of Instances (num. nodes in brackets) for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 6	HH 6	L2F 6	RR 6
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	$0.968\,61 \pm 0.005\,79$
	F_MPS.json	0.8072 ± 0.0736	0.7659 ± 0.0998	0.8996 ± 0.0279	0.9219 ± 0.0671
	I_MPS.json	$0.964\,70 \pm 0.008\,99$	0.8188 ± 0.0968	$0.943\,15 \pm 0.009\,16$	0.9435 ± 0.0358
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	0.9120 ± 0.0290	$0.888\,06 \pm 0.008\,75$	0.9142 ± 0.0141	0.9394 ± 0.0129
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	0.9252 ± 0.0155
	F_PP.json	0.9204 ± 0.0150	0.8931 ± 0.0280	0.8291 ± 0.0257	0.8974 ± 0.0183
	I_PP.json	0.9437 ± 0.0139	$0.922\,44 \pm 0.009\,38$	0.8836 ± 0.0206	0.9156 ± 0.0185
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	0.9327 ± 0.0155	$0.897\,64 \pm 0.009\,56$	0.8523 ± 0.0235	0.9066 ± 0.0156
SV	FA_SV_noOpt.json	0.9038 ± 0.0246	-	-	0.9581 ± 0.0133
	FA_SV_opt.json	0.9416 ± 0.0157	-	-	0.9620 ± 0.0140
	F_SV.json	-	-	-	-
	I_SV.json	0.9443 ± 0.0157	0.9360 ± 0.0135	-	0.9220 ± 0.0539
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.9319 ± 0.0168	0.9124 ± 0.0220	-	0.9429 ± 0.0216
	TS_SV.json	0.9444 ± 0.0193	0.9314 ± 0.0129	-	0.9141 ± 0.0345

Table 12: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 6$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

7 Tables for depth $P = 7$

	graph_type depth	ER 7	HH 7	L2F 7	RR 7
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	-
	F_MPS.json	10 [40]	31 [39–144]	10 [40]	21 [40–100]
	L_MPS.json	10 [40]	31 [39–144]	10 [40]	21 [40–100]
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	-	-	-
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	-
	F_PP.json	10 [40]	31 [39–144]	10 [40]	21 [40–100]
	L_PP.json	10 [40]	31 [39–144]	10 [40]	21 [40–100]
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	-	-	-
SV	FA_SV_noOpt.json	11 [11–17]	-	-	42 [10–20]
	FA_SV_opt.json	11 [11–17]	-	-	42 [10–20]
	F_SV.json	-	-	-	-
	L_SV.json	32 [10–17]	10 [12]	-	165 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	10 [12]	-	131 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	130 [10–20]

Table 13: **Number of Instances (num. nodes in brackets) for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

graph.type depth		ER 7	HH 7	L2F 7	RR 7
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	-
	F_MPS.json	0.8170 ± 0.0738	0.776 ± 0.101	0.8783 ± 0.0282	0.9102 ± 0.0756
	I_MPS.json	0.97193 ± 0.00935	0.8374 ± 0.0935	0.95229 ± 0.00786	0.9510 ± 0.0388
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	-	-	-
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	-
	F_PP.json	0.9031 ± 0.0384	0.8987 ± 0.0270	0.8475 ± 0.0201	0.9011 ± 0.0184
	I_PP.json	0.9491 ± 0.0139	0.93143 ± 0.00901	0.8955 ± 0.0236	0.9153 ± 0.0196
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	-	-	-
SV	FA_SV_noOpt.json	0.9084 ± 0.0296	-	-	0.9671 ± 0.0117
	FA_SV_opt.json	0.9514 ± 0.0164	-	-	0.9717 ± 0.0125
	F_SV.json	-	-	-	-
	I_SV.json	0.9535 ± 0.0150	0.9425 ± 0.0125	-	0.9229 ± 0.0595
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.9422 ± 0.0148	0.9177 ± 0.0224	-	0.9514 ± 0.0178
	TS_SV.json	0.9524 ± 0.0198	0.9384 ± 0.0104	-	0.9221 ± 0.0339

Table 14: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 7$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

8 Tables for depth $P = 8$

	graph_type depth	ER 8	HH 8	L2F 8	RR 8
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	10 [40]
	F_MPS.json	10 [40]	31 [39–144]	10 [40]	21 [40–100]
	L_MPS.json	10 [40]	31 [39–144]	10 [40]	21 [40–100]
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	10 [40]	10 [39]	10 [40]	20 [40]
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	10 [40]
	F_PP.json	10 [40]	31 [39–144]	10 [40]	21 [40–100]
	L_PP.json	10 [40]	31 [39–144]	10 [40]	21 [40–100]
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	10 [40]	10 [39]	10 [40]	20 [40]
SV	FA_SV_noOpt.json	11 [11–17]	-	-	43 [10–20]
	FA_SV_opt.json	11 [11–17]	-	-	43 [10–20]
	F_SV.json	-	-	-	-
	L_SV.json	32 [10–17]	10 [12]	-	165 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	10 [12]	-	132 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	131 [10–20]

Table 15: **Number of Instances (num. nodes in brackets) for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 8	HH 8	L2F 8	RR 8
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	0.98288 ± 0.00287
	F_MPS.json	0.8165 ± 0.0404	0.768 ± 0.102	0.8483 ± 0.0935	0.9383 ± 0.0627
	I_MPS.json	0.9761 ± 0.0103	0.843 ± 0.101	0.96014 ± 0.00680	0.9606 ± 0.0320
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	0.9380 ± 0.0283	0.90313 ± 0.00980	0.9140 ± 0.0181	0.96415 ± 0.00992
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	0.9341 ± 0.0157
	F_PP.json	0.9182 ± 0.0226	0.9011 ± 0.0304	0.8523 ± 0.0206	0.9044 ± 0.0218
	I_PP.json	0.9529 ± 0.0137	0.93873 ± 0.00854	0.9015 ± 0.0239	0.9185 ± 0.0211
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	0.9363 ± 0.0148	0.91134 ± 0.00720	0.8703 ± 0.0218	0.9197 ± 0.0146
SV	FA_SV_noOpt.json	0.9160 ± 0.0323	-	-	0.9733 ± 0.0108
	FA_SV_opt.json	0.9606 ± 0.0153	-	-	0.9790 ± 0.0116
	F_SV.json	-	-	-	-
	I_SV.json	0.9598 ± 0.0141	0.9470 ± 0.0111	-	0.9261 ± 0.0601
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.9467 ± 0.0147	0.9234 ± 0.0184	-	0.9570 ± 0.0166
	TS_SV.json	0.9592 ± 0.0195	0.94300 ± 0.00962	-	0.9285 ± 0.0334

Table 16: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 8$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

9 Tables for depth $P = 9$

	graph_type depth	ER 9	HH 9	L2F 9	RR 9
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	-
	F_MPS.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
	L_MPS.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	-	-	-
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	-
	F_PP.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
	L_PP.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	-	-	-
SV	FA_SV_noOpt.json	11 [11–17]	-	-	42 [10–20]
	FA_SV_opt.json	11 [11–17]	-	-	42 [10–20]
	F_SV.json	-	-	-	-
	L_SV.json	32 [10–17]	10 [12]	-	165 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	10 [12]	-	131 [10–20]
	TS_SV.json	32 [10–17]	10 [12]	-	130 [10–20]

Table 17: **Number of Instances (num. nodes in brackets) for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 9	HH 9	L2F 9	RR 9
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	-
	F_MPS.json	0.780	0.7833 ± 0.0972	0.891	0.9268 ± 0.0735
	I_MPS.json	0.986	0.8618 ± 0.0927	0.962	0.9308 ± 0.0449
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	-	-	-	-
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	-
	F_PP.json	0.908	0.9064 ± 0.0318	0.845	0.9104 ± 0.0261
	I_PP.json	0.964	0.94445 ± 0.00803	0.872	0.9078 ± 0.0226
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	-	-	-	-
SV	FA_SV_noOpt.json	0.9190 ± 0.0350	-	-	0.9774 ± 0.0105
	FA_SV_opt.json	0.9667 ± 0.0158	-	-	0.9836 ± 0.0108
	F_SV.json	-	-	-	-
	I_SV.json	0.9639 ± 0.0133	0.9478 ± 0.0124	-	0.9246 ± 0.0667
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.9531 ± 0.0145	0.9296 ± 0.0161	-	0.9620 ± 0.0170
	TS_SV.json	0.9647 ± 0.0189	0.94662 ± 0.00897	-	0.9313 ± 0.0326

Table 18: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 9$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

10 Tables for depth $P = 10$

	graph_type depth	ER 10	HH 10	L2F 10	RR 10
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	11 [40–100]
	F_MPS.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
	L_MPS.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	11 [40–100]
	F_PP.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
	L_PP.json	1 [40]	31 [39–144]	1 [40]	11 [40–100]
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	1 [40]	30 [39–144]	1 [40]	11 [40–100]
SV	FA_SV_noOpt.json	11 [11–17]	-	-	26 [10–24]
	FA_SV_opt.json	11 [11–17]	-	-	25 [10–22]
	F_SV.json	-	-	-	-
	L_SV.json	32 [10–17]	10 [12]	-	140 [10–20]
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	32 [10–17]	9 [12]	-	139 [10–20]
	TS_SV.json	31 [10–17]	9 [12]	-	136 [10–19]

Table 19: **Number of Instances (num. nodes in brackets) for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).

	graph.type depth	ER 10	HH 10	L2F 10	RR 10
MPS	FA_MPS_no_opt.json	-	-	-	-
	FA_MPS_opt.json	-	-	-	$0.986\,01 \pm 0.009\,18$
	F_MPS.json	0.828	0.774 ± 0.109	0.890	0.9280 ± 0.0900
	I_MPS.json	0.988	0.866 ± 0.101	0.965	0.9383 ± 0.0455
	RTS_MPS.json	-	-	-	-
	TQA_MPS_no_opt.json	-	-	-	-
	TQA_MPS_opt.json	0.952	0.8075 ± 0.0930	0.916	0.9655 ± 0.0173
PP	FA_PP_no_opt.json	-	-	-	-
	FA_PP_opt.json	-	-	-	0.9364 ± 0.0158
	F_PP.json	0.891	0.9110 ± 0.0325	0.848	0.9127 ± 0.0258
	I_PP.json	0.966	$0.949\,10 \pm 0.007\,64$	0.875	0.9011 ± 0.0244
	RTS_PP.json	-	-	-	-
	TQA_PP_no_opt.json	-	-	-	-
	TQA_PP_opt.json	0.949	$0.917\,75 \pm 0.006\,73$	0.835	0.9225 ± 0.0158
SV	FA_SV_noOpt.json	0.9223 ± 0.0358	-	-	$0.978\,18 \pm 0.009\,62$
	FA_SV_opt.json	0.9708 ± 0.0159	-	-	$0.985\,20 \pm 0.009\,14$
	F_SV.json	-	-	-	-
	I_SV.json	0.9671 ± 0.0126	0.9484 ± 0.0111	-	0.9399 ± 0.0605
	TQA_SV_noOpt.json	-	-	-	-
	TQA_SV_opt.json	0.9576 ± 0.0143	0.9322 ± 0.0206	-	0.9679 ± 0.0150
	TS_SV.json	0.9696 ± 0.0188	$0.951\,90 \pm 0.008\,32$	-	0.9426 ± 0.0314

Table 20: **Avg. Approx. Ratio $\pm$ standard deviation for $P = 10$.** Graph types are Erdos Renyi (ER), Heavy-Hex (HH), Line-to-Full (L2F), and Random Regular (RR).